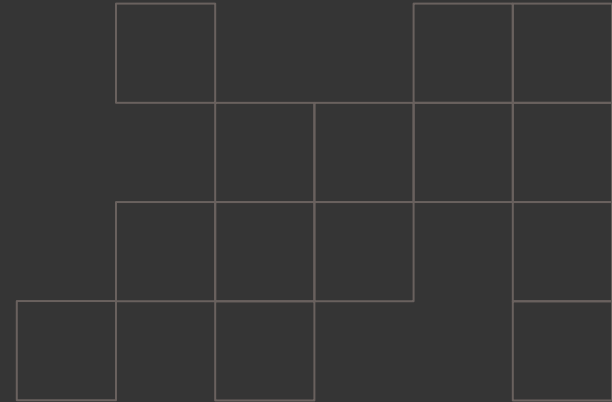


Ricardo Gaeta

Humanities Hub

Discover local arts and cultural institutions



Contents

1. ***Introduction***
2. ***Overview Diagrams***
3. ***Front-end Implementation***
4. ***Back-end Implementation***
5. ***Future Potential***
6. ***Live Demo***

Introduction

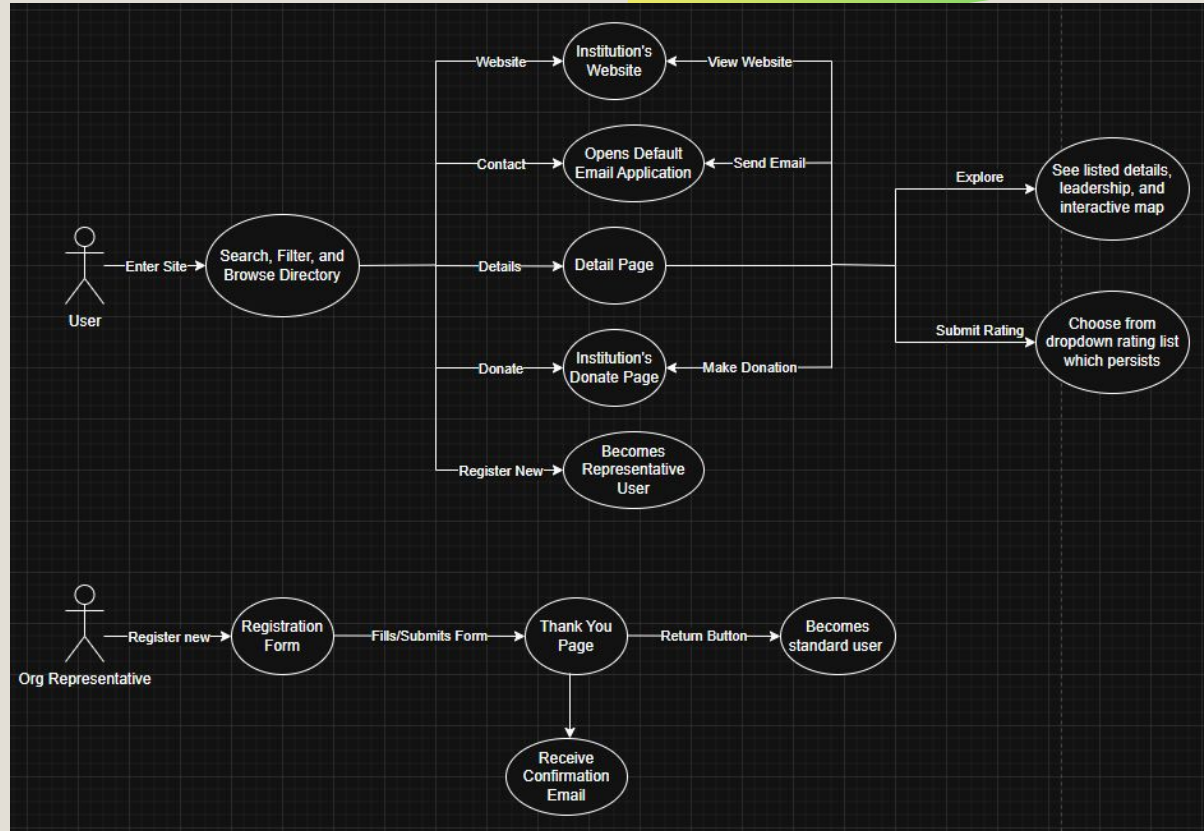
- Goal
 - Easy to use website to discover, learn about, and support Humanities based businesses/organizations.
 - Utilize search, filters, and interactive maps to find locations/institutions.
 - Utilize a registration form to register and share institutions with others.
- Why?
 - To show that tech can be used to promote and spread cultural/natural/artistic experiences
 - My own interest with these types of locations





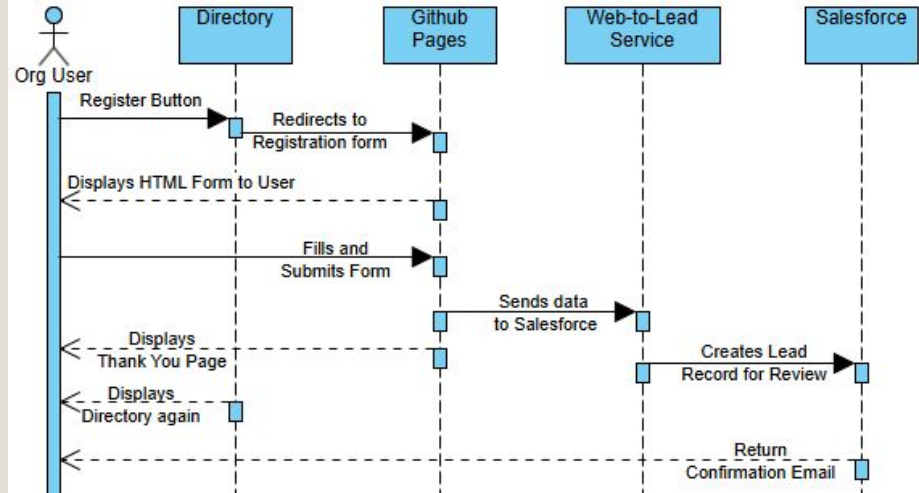
Overview

Use-Case Diagram

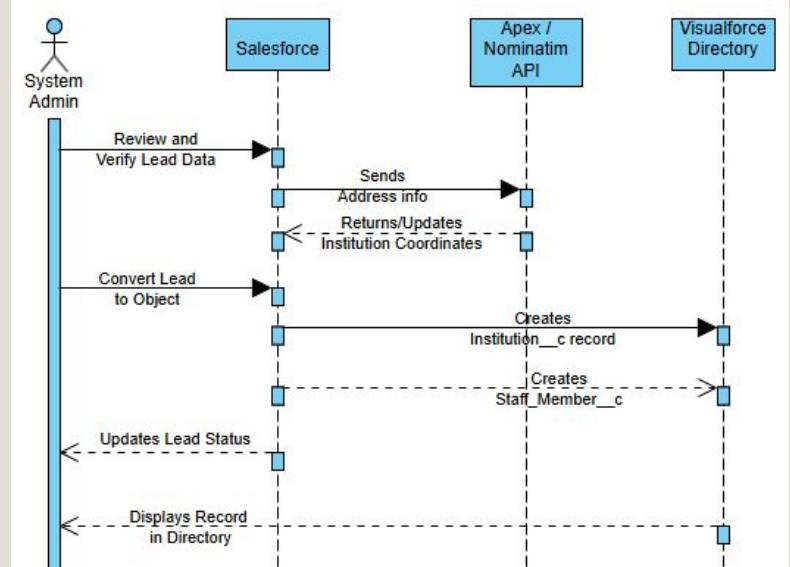


Sequence Diagram

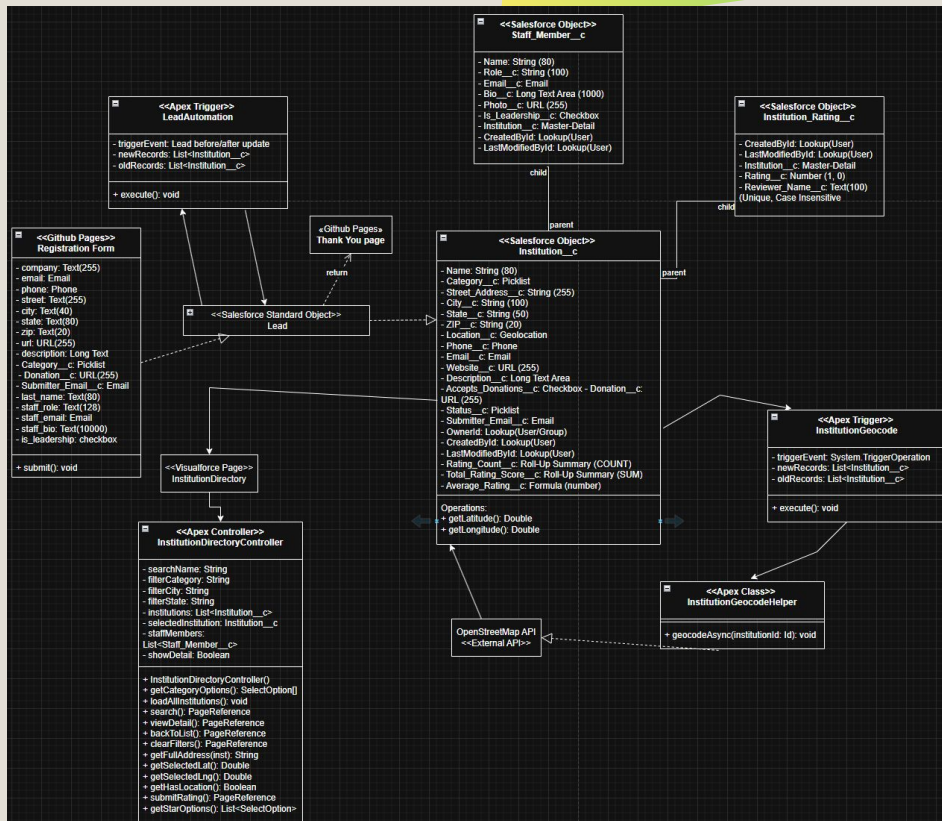
- Specifically for Registration

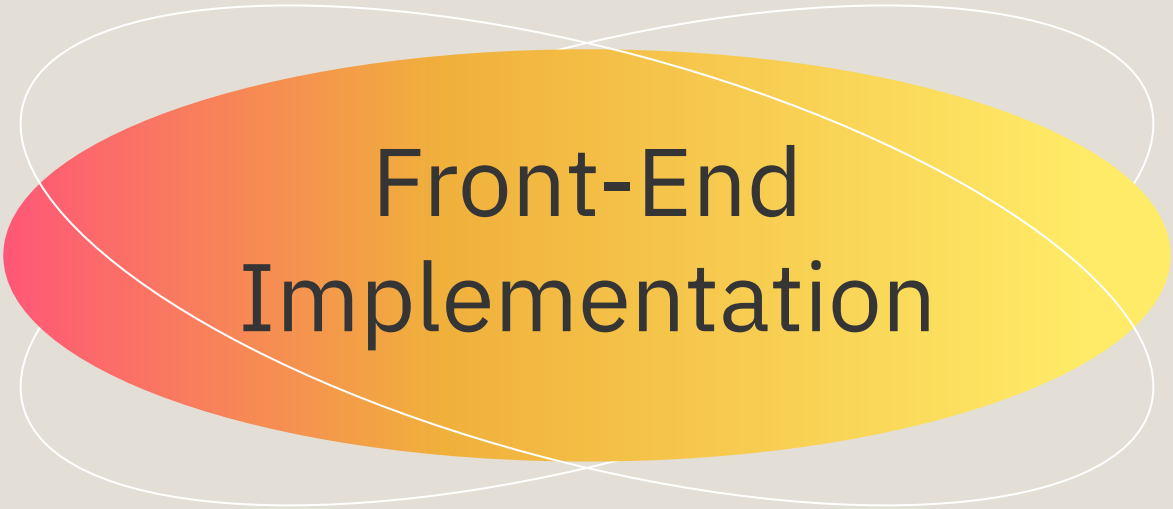


Public User Sequence



System Admin Sequence

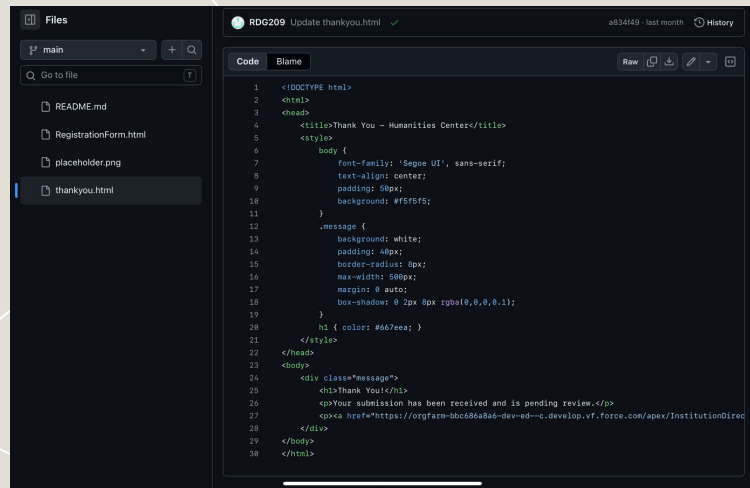
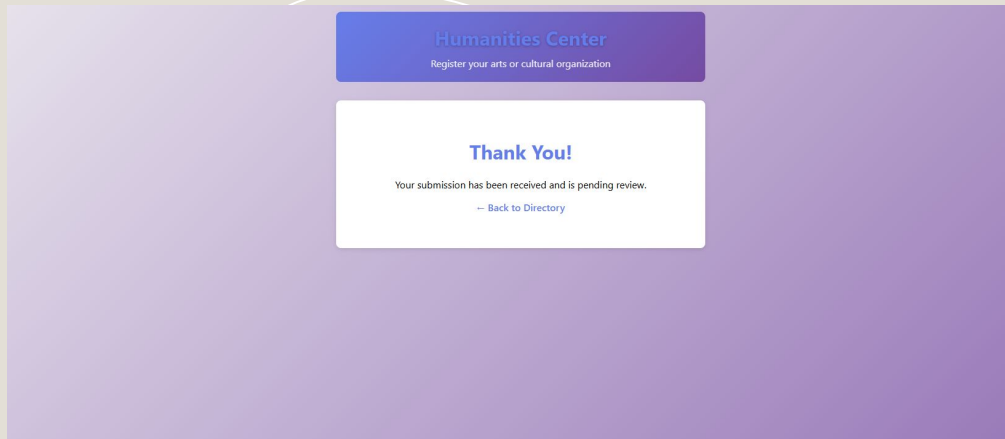


A graphic featuring a horizontal oval with a color gradient from red on the left to yellow on the right. The text "Front-End Implementation" is centered within this oval. Two thin, white, hand-drawn style ellipses are positioned behind the colored oval, one above and one below, creating a layered effect.

Front-End Implementation

Github Pages

- Thank you Page
 - ◆ html/css page with a redirect
 - ◆ Hosted on Github
 - ◆ Redirects to visualforce webpage



Github Pages

→ Registration Form

- ◆ HTML/CSS form, decorated with same gradient theme as website
- ◆ Directly gathers data then POSTs to web-to-lead endpoint
- ◆ Creates Lead in salesforce backend

Humanities Center
Register your arts or cultural organization

Organization Name *
e.g.: Dunder Mifflin Inc.

Phone
e.g.: ##### or ###-###-#### or ####.####

Institution Contact Email *
e.g.: name@gmail.com

Organization Description
Tell us about your organization's mission and activities...

☒ **This organization accepts donations**

Donation Page URL
https://www.example.com/donate

Street Address

```

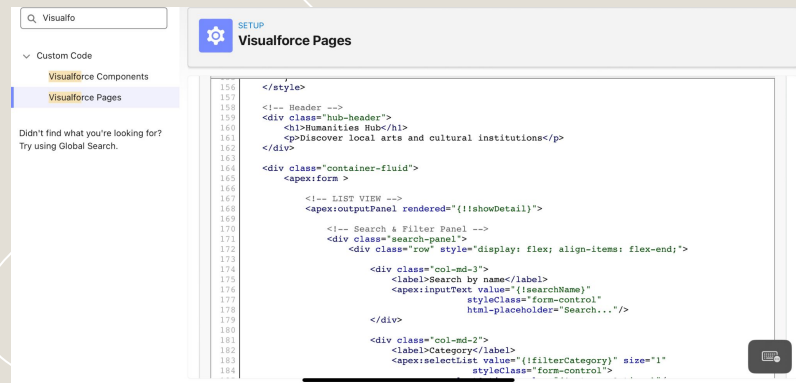
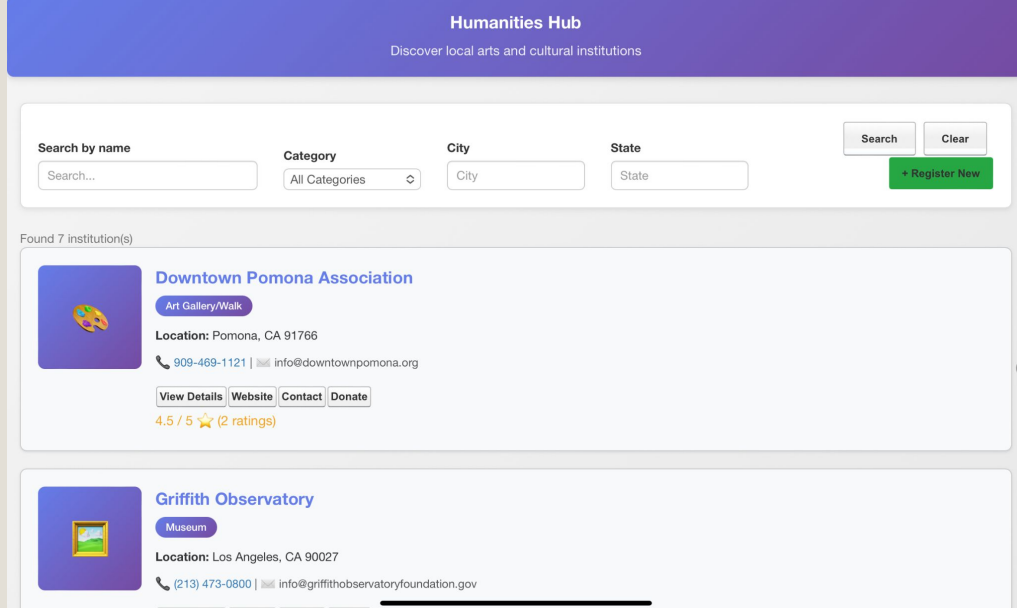
RDG209.github.io / RegistrationForm.html
Code Blame
<form action="https://webto.salesforce.com/servlet/servlet.WebToLead?encoding=UTF-8&orgId=
91
92
93
94 <input type="hidden" name="oid" value="00G5000000572T">
95 <input type="hidden" name="leadSource" value="Web">
96 <input type="hidden" name="first_name" value="">
97 <input type="hidden" name="retURL" value="https://RDG209.github.io/thankyou.html">
98 <input type="hidden" name="debug" value="0">
99 <input type="hidden" name="debugEmail" value="rgardner@csuoh.edu">
100 <input type="hidden" name="country" value="United States">
101 <input type="hidden" name="owner_id" value="00G5000000353h">
102 <input type="hidden" name="lead_source" value="Website">
103
104 <!-- Primary Org Info -->
105
106 <label for="company">Organization Name <span class="required">*</span></label>
107 <input id="company" maxlength="48" name="company" size="28" type="text" required placeh
108
109 <label for="phone">Phone</label>
110 <input id="phone" maxlength="48" name="phone" size="28" type="tel" placeholder="e.g.: #
111
112 <label for="email">Institution Contact Email <span class="required">*</span></label>
113 <input id="email" maxlength="88" name="email" size="28" type="email" required placeho
114
115 <!-- Description -->
116 <label for="description">Organization Description</label>
117 <textarea id="description" name="description" rows="4"
118   placeholder="Tell us about your organization's mission and activities..."></textare
119
120 <!-- Accepts Donations -->
121 <label>
122   <input type="checkbox" name="Accepts_Donations_c" value="1" checked/>
123   This organization accepts donations
124 </label>
125
  
```

Visualforce Page

→ Salesforce-Powered Directory

◆ Utilizes Apex code

- HTML is nearly unchanged other than basic syntax and field names
- CSS is effectively normal
- JS-like `<script>` field for the leaflet.js map





Back-End Implementation

Apex Classes

- Most similar to Java code

→ InstitutionDirectoryController

- ◆ Primary class controlling the directory
- ◆ Handles filters, ratings, search, etc

```

Apex Class  Version Settings
153  filterCity = '';
154  filterState = '';
155  loadAllInstitutions();
156  return null;
157  }
158  // Map methods for coords
159  public Double getSelectedLat() {
160  if (selectedInstitution == null || selectedInstitution.Location__c == null) {
161  return 34.0522; // Default: Los Angeles
162  }
163  return selectedInstitution.Location__c.getLatitude();
164  }
165  }
166  public Double getSelectedLong() {
167  if (selectedInstitution == null || selectedInstitution.Location__c == null) {
168  return -118.2437; // Default: Los Angeles
169  }
170  return selectedInstitution.Location__c.getLongitude();
171  }
172  }
173  public Boolean getHasLocation() {
174  return selectedInstitution != null && selectedInstitution.Location__c != null;
175  }
176  }
177  // Submit a rating
178  public PageReference submitting() {
179  if (selectedInstitution == null) return null;
180  }
181  if (selectedRating == null || selectedRating < 1 || selectedRating > 5) {
182  ApexPages.addMessage(new ApexPages.Message(

```

→ InstitutionGeocodeHelper

- ◆ Helper class for geocode automation

```

Apex Class  Version Settings
1  public class InstitutionGeocodeHelper {
2
3  @future(callout=true)
4  public static void geocodeAsync(Id institutionId) {
5
6  // Re-query the record - include latitude/longitude fields
7  Institution__c inst = [
8  SELECT Id, Street_Address__c, City__c, State__c, ZIP__c,
9  Location_Latitude__s, Location_Longitude__s
10 FROM Institution__c
11 WHERE Id = :institutionId
12 LIMIT 1
13 ];
14
15 // Build address string
16 List<String> addressParts = new List<String>();
17 if (String.isNotBlank(inst.Street_Address__c)) addressParts.add(inst.Street_Address__c);
18 if (String.isNotBlank(inst.City__c)) addressParts.add(inst.City__c);
19 if (String.isNotBlank(inst.State__c)) addressParts.add(inst.State__c);
20 if (String.isNotBlank(inst.ZIP__c)) addressParts.add(inst.ZIP__c);
21
22 String fullAddress = String.join(addressParts, ', ');
23 String encodedAddress = EncodingUtil.urlEncode(fullAddress, 'UTF-8');
24
25 // Call OpenStreetMap Nominatim
26 http http = new Http();
27 httprequest request = new HttpRequest();
28 request.setEndpoint('https://nominatim.openstreetmap.org/search?format=json&' + encodedAddress);
29 request.setMethod('GET');
30 request.setHeader('User-Agent', 'HumanitiesHub/1.0 (you@gmail.com)');

```

Apex Triggers

- Most similar to Java code

→ InstitutionGeocode

- ◆ Triggers on lead creation
- ◆ Generates the latitude/longitude values

→ LeadAutomation

- ◆ Triggers on register form submission
- ◆ Creates the lead and notifies the user (notification does not trigger)

```

Apex Trigger | Version Settings
is active ☒

1 trigger InstitutionGeocode on Institution__c (after insert, after update) {
2     List<Id> toGeocode = new List<Id>();
3
4     for (Institution__c inst : Trigger.new) {
5
6         Boolean hasAddress = String.isNotBlank(inst.Street_Address__c)
7             && String.isNotBlank(inst.City__c)
8             && String.isNotBlank(inst.State__c);
9
10        Boolean locationEmpty = inst.Location__c == null;
11
12        Boolean addressChanged = false;
13        if (Trigger.isUpdate) {
14            Institution__c old = Trigger.oldMap.get(inst.Id);
15            addressChanged = (inst.Street_Address__c != old.Street_Address__c)
16                || (inst.City__c != old.City__c)
17                || (inst.State__c != old.State__c)
18                || (inst.Zip__c != old.Zip__c);
19        }
20
21        if (hasAddress && (Trigger.isInsert || locationEmpty || addressChanged)) {
22            toGeocode.add(inst.Id);
23        }
24    }
25
26    // Call future method once per record outside the loop
27    for (Id instId : toGeocode) {
28        InstitutionGeocodeHelper.geocodeAsync(instId);
29    }
30
Position: Ln 31, Ch 2      Total: Ln 31, Ch 1130

```

```

Apex Trigger | Version Settings
is active ☒

1 trigger LeadAutomation on Lead (before insert, after insert) {
2     // BEFORE INSERT: Set owner
3     if (Trigger.isBefore && Trigger.isInsert) {
4         for (Lead l : Trigger.new) {
5             if (l.LeadSource == 'Website' || l.LeadSource == 'Web') {
6                 l.OwnerId = '00sg000000151h';
7             }
8         }
9     }
10
11    // AFTER INSERT: Send email (can't modify records here)
12    if (Trigger.isAfter && Trigger.isInsert) {
13        List<Messaging.SingleEmailMessage> emails = new List<Messaging.SingleEmailMessage>();
14
15        for (Lead l : Trigger.new) {
16            if (l.LeadSource == 'Website' && l.Submitter_Email__c != null) {
17                Messaging.SingleEmailMessage mail = new Messaging.SingleEmailMessage();
18                mail.setToAddresses(new String[] { l.Submitter_Email__c });
19                mail.setSubject('Humanities Hub Registration Received');
20                mail.setPlainTextBody(
21                    'Dear ' + l.LastName + ',\n\n' +
22                    'Thank you for registering ' + l.Company + ' with Humanities Hub.\n\n' +
23                    'Your submission is under review. We will notify you when your organization appears in our directory.\n\n' +
24                    '- Humanities Hub Team'
25                );
26                emails.add(mail);
27            }
28        }
29    }
30
Position: Ln 39, Ch 2      Total: Ln 39, Ch 1037

```



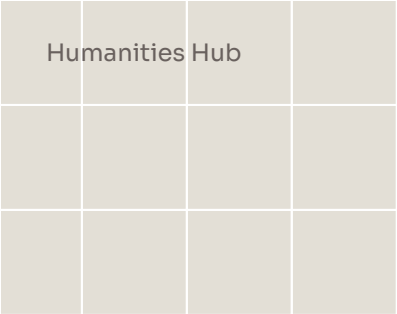
Future Potential

Future Potentials

- Problems to fix
 - ◆ Additional Functionalities
 - Social Systems, Permanent Accounts, Full App deployment
- Beneficial Potential
 - ◆ Contracting of the 'company'/service to groups like national parks/ state cultural sites



DEMO



Thank You!

Any Questions?

References:

[1] *Flowchart Maker & Online Diagram Software*. <https://app.diagrams.net/>. Accessed 21 Apr. 2026.

[2] “VP Online.” *Online Drawing Tool*, <https://online.visual-paradigm.com/app/diagrams>. Accessed 21 Apr. 2026.

[3] *Salesforce. Salesforce Developers*. https://developer.salesforce.com/docs/atlas.en-us.api.meta/api/sforce_api_quickstart_intro.htm. Accessed 30 Apr. 2026.

[4] “Developer Portal.” *Developer*, <https://developer.salesforce.com/>. Accessed 30 Apr. 2026.

